

RESTAURANT MANAGEMENT SYSTEM (RestMS)

PROJECT PROPOSAL DOCUMENT – PREMIUM VERSION

1. PROPOSAL SUMMARY

We propose to develop a **Premium Restaurant Management System (RestMS)** — a complete digital ecosystem with enterprise POS, mobile apps (customer + staff), AI-powered operations, multi-branch management, franchise tools, and full delivery platform integration. Designed for **large restaurant chains and food brands (15+ outlets)** requiring enterprise-grade scalability, automation, and performance.

Ideal For: Large restaurant chains, food franchises, hotel F&B; departments, food delivery brands

2. OBJECTIVES

- Fully automate all restaurant operations end-to-end across all branches
- Provide seamless digital experience for customers, kitchen, and management
- Enable real-time multi-branch visibility and centralized control
- Implement AI-driven menu optimization, demand forecasting, and upselling
- Ensure enterprise-grade security, compliance, and scalability

3. SCOPE OF WORK

3.1 Website + Customer Portal (Enterprise)

- Premium animated restaurant website Customer
- self-service ordering portal AI-powered
- personalized menu recommendations Live order
- tracking with map view QR code dine-in ordering
- (scan & order) Multi-outlet selection and delivery
- zones Blog, chef stories & event promotion section
-

Cost: Rs. 90,000 - Rs. 1,40,000

3.2 Enterprise POS & Billing Suite

- Multi-terminal cloud POS system
- Offline mode with auto-sync
- Dynamic pricing (happy hours, seasonal)

- Multi-currency and multi-tax support
- E-invoice and GST filing ready
- Cashier-wise shift reports and cash drawer management

Cost: Rs. 1,00,000 - Rs. 1,70,000

3.3 Multi-Branch & Franchise Management

- Centralized dashboard for all branches
- Branch-wise P&L; and performance comparison
- Franchise royalty and fee management
- Centralized menu with branch-level customization
- Inter-branch transfer and stock management

Cost: Rs. 1,00,000 - Rs. 1,60,000

3.4 Kitchen Production & Smart Operations

- Multi-station KDS with prep timer
- Recipe management with auto costing
- AI-based demand forecasting for prep planning
- Smart wastage reduction alerts
- Equipment maintenance scheduling

Cost: Rs. 70,000 - Rs. 1,20,000

3.5 Advanced Inventory & Supply Chain

- Centralized multi-branch inventory
- Auto purchase order generation
- Supplier portal for order management
- Real-time food cost and margin tracking
- Shelf-life and expiry management

Cost: Rs. 80,000 - Rs. 1,30,000

3.6 Staff Management + HRMS

- Complete HRMS with payroll integration
- Face recognition attendance
- Shift scheduling and auto-rostering
- Performance KPIs and incentive management
- Training and certification tracking

Cost: Rs. 80,000 - Rs. 1,30,000

3.7 Mobile Applications (Android / iOS)

- Customer App: Menu browsing, ordering, live tracking, loyalty rewards, table booking
- Staff App: Order management, KDS view, attendance, daily reports, shift alerts

Cost: Rs. 2,00,000 - Rs. 3,50,000

4. ADDITIONAL FEATURES

4.1 AI Menu Optimization & Upselling

- Personalized item recommendations
 - Peak hour demand prediction
 - Menu profitability scoring & optimization

Cost: Rs. 80,000 - Rs. 1,20,000

4.2 Delivery Platform Integration (All-in-One)

- Zomato, Swiggy, ONDC, own delivery integration
 - Centralized order hub across all platforms
 - Delivery partner management

Cost: Rs. 60,000 - Rs. 1,00,000

4.3 WhatsApp + SMS + Email + Push Notifications

- Full CRM-based customer communication
- Automated campaign scheduler
- Personalized offer delivery

Cost: Rs. 40,000 - Rs. 65,000

4.4 Customer CRM & Advanced Loyalty Engine

- Tiered loyalty program (Bronze/Silver/Gold)
- Gamified rewards & referral system
- Customer lifetime value tracking

Cost: Rs. 50,000 - Rs. 80,000

4.5 Security, Audit & Compliance

- Role-based access with multi-level approvals
- Full transaction audit trail AES-
- 256 data encryption FSSAI &
- GST compliance tools

Cost: Rs. 50,000 - Rs. 80,000

5. TECHNOLOGY STACK (MERN + AI + MOBILE)

- **Frontend:** React.js
- **Backend:** Node.js + Express.js
- **Database:** MongoDB + Redis (caching)
- **Mobile Apps:** React Native / Flutter
- **AI/ML:** Python-based demand forecasting & recommendation engine
- **Integrations:** Zomato/Swiggy/ONDC, WhatsApp Business API, Razorpay/PayU

- **Hosting:** AWS (EC2, S3, CloudFront, RDS)

6. TOTAL PROJECT COSTING

Development Cost (One-time)	Rs. 9,00,000 - Rs. 16,00,000+ (One-time)
Infrastructure	AWS Hosting: Rs. 10,000 - Rs. 50,000/month
Infrastructure	Domain + SSL: Rs. 2,000/year
Infrastructure	CDN & Backup: Rs. 15,000/year
Infrastructure	API Integrations: Rs. 5,000 - Rs. 20,000/month
Maintenance Cost	Rs. 50,000 - Rs. 1,00,000/month/month

7. PROJECT TIMELINE

Phase	Duration
Planning	2 Weeks
Development	12-16 Weeks
Testing	2-3 Weeks
Deployment	1-2 Weeks
Total Duration	4-5 Months

8. LIMITATIONS

- High initial investment
- Requires dedicated IT support team
- Third-party API costs are recurring
- Staff training mandatory for smooth adoption

9. FUTURE UPGRADES

- IoT smart kitchen equipment integration
- AI chatbot for 24/7 customer support
- Blockchain-based supply chain tracking
- Augmented Reality (AR) digital menu

10. SUPPORT AND WARRANTY

- 3 months free support
- Priority bug fixing
- Dedicated account manager & support team

11. TERMS AND CONDITIONS

- 40% advance payment
- 30% mid-phase
- 30% after completion
- Additional modules charged separately

12. CONCLUSION

This Premium RestMS solution provides a **complete digital transformation platform** for restaurant chains — combining enterprise POS, AI-powered operations, mobile apps, franchise tools, delivery integrations, and full analytics, ensuring long-term growth, brand consistency, and operational excellence.

13. CONTACT DETAILS

Company Name:	Kartsho Digital
Parent Company:	A Venture of Kartsho Enterprises
Phone:	+91 9453134901 / +91 9528660578
Email:	kartshoenterprises@gmail.com
Website:	digital.kartsho.com
Address:	Godhoopatti, Patti, Pratapgarh, Uttar Pradesh - 230134